GeorgeFisher_HSE711_Final_Project

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```
library(tidyverse)
```

```
# Read in both data sets
count.df <- read.csv(file = "data/counts.csv", header = TRUE)
meta_data.df <- read.csv(file = "data/meta_data.csv", header = TRUE)
```

```
gene_ensembl_map <- tibble(
  gene_symbols = c(
    "BRCA1",
    "BRCA2",
    "PALB2",
    "BARD1",
    "RAD51C",
    "RAD51D",
    "BRIP1",
    "ATM",
    "CHEK2",
    "CDK12"
  ),
  ensembl.ids = c(
    "ENSG00000012048",
    "ENSG00000139618",
    "ENSG00000083093",
    "ENSG00000138376",
    "ENSG00000108384",
    "ENSG00000185379",
    "ENSG00000136492",
    "ENSG00000149311",
    "ENSG00000183765",
    "ENSG00000167258"
  )
)
```

```
# Rename the first column of the counts data
names(count.df)[1] <- "gene_symbols"

# Store the sample identifiers from the count.df
sample_ids <- names(count.df)[-1]
```

```
# Remove the version suffix of the Ensembl identifier
count.df <- count.df %>%
  mutate(
    ensembl.ids = str_remove(gene_symbols, "\\..*$")
  )
```

```
heatmap_counts <- count.df %>%
  filter(
    ensembl.ids %in% gene_ensembl_map$ensembl.ids
    ) %>%
  select(
    ensembl.ids,
    all_of(sample_ids)
  ) %>%
    arrange(
      match(
        ensembl.ids,
        gene_ensembl_map$ensembl.ids
      )
)
```

```
# Verify that all counts_data sample identifiers are present in the barcode meta_data field
all(sample_ids %in% meta_data.df$barcode)
```

```
## [1] FALSE
```

```
count.matrix <- as.matrix(
  heatmap_counts %>%
    select(all_of(sample_ids))
)
```

```
rownames(count.matrix) <- gene_ensembl_map$gene_symbols[
  match(
    heatmap_counts$ensembl.ids,
    gene_ensembl_map$ensembl.ids
  )
]
```

```
meta_data.df <- meta_data.df %>%
  # The barcode variable will eventually be used to join the metadata to the count data
  # The hyphen in the barcode variable needs to be changed to a dot
  mutate(barcode = str_replace_all(barcode, "-", "."))
```

```
heatmap_metadata <- meta_data.df %>%
  select(
    barcode,
    PAM50_Subtype = paper_BRCA_Subtype_PAM50
  ) %>%
  filter(
    barcode %in% colnames(count.matrix),
    !is.na(PAM50_Subtype)
  ) %>%
  distinct(
    barcode,
    .keep_all = TRUE
  ) %>%
  mutate(
    PAM50_Subtype = factor(
      PAM50_Subtype,
      levels = c(
        "Basal",
```

```
        "Her2",
        "LumA",
        "LumB",
        "Normal"
      ),
      labels = c(
        "Basal",
        "Her2-enriched",
        "Luminal A",
        "Luminal B",
        "Normal"
      )
    )
  ) %>%
  arrange(PAM50_Subtype)
```

```
library(ComplexHeatmap)
library(circlize)
```

```
count.matrix <- count.matrix[
  ,
  heatmap_metadata$barcode,
  drop = FALSE
]

class(count.matrix) <- "numeric" # enforce numeric mode if needed

log2_count_matrix <- log2(count.matrix)
```

```
# standardize each gene across samples using row Z-scores
scaled_count_matrix <- t(
  scale(
    t(log2_count_matrix)
  )
)
```

```
# standardize each gene across samples using row Z-scores
scaled_count_matrix <- t(
  scale(
    t(log2_count_matrix)
  )
)
```

```
subtype_colors <- c(
      "Basal" =          "#D55E00",
      "HER2-Enriched" = "#CC79A7",
      "Luminal A" =      "#009E73",
      "Luminal B" =      "#E69F00",
      "Normal" =         "#5684E9"
      )
```

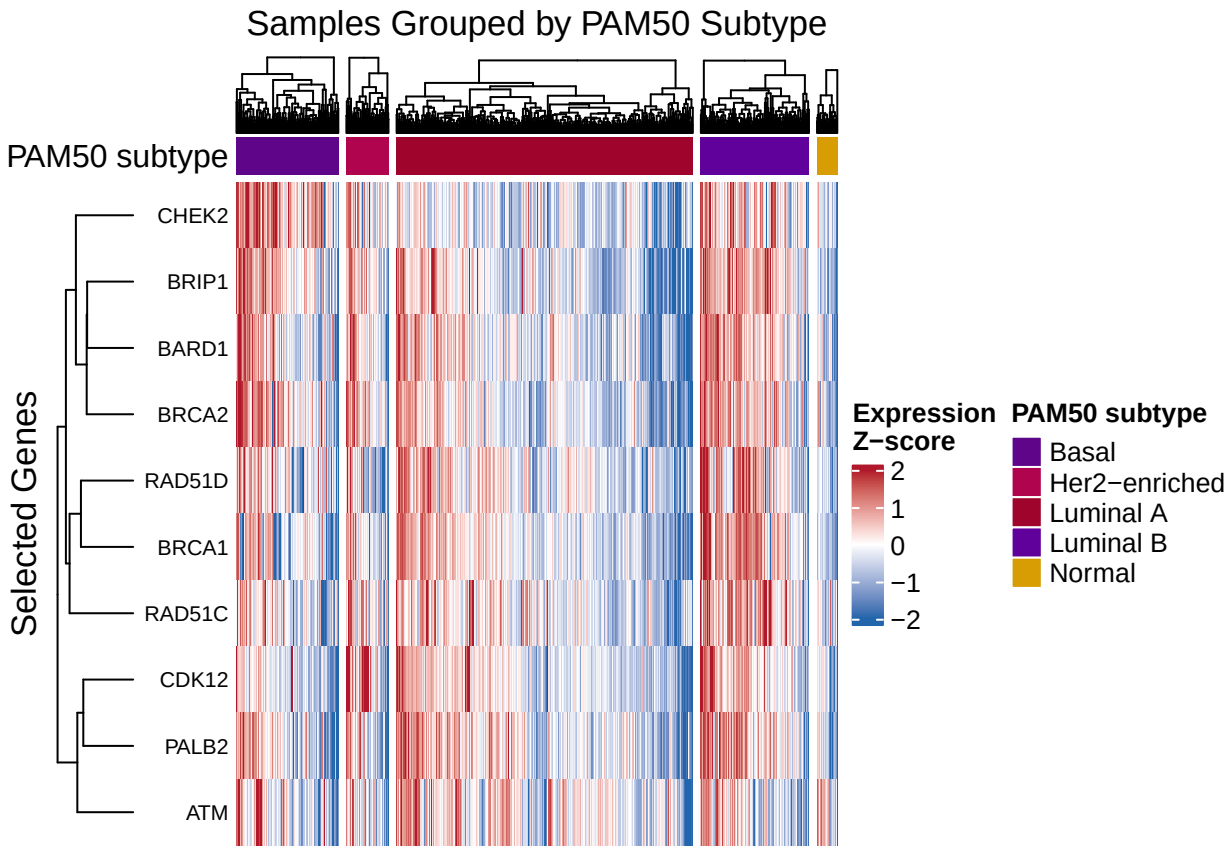
```
subtype_annotation <- HeatmapAnnotation(
  `PAM50 subtype` = heatmap_metadata$PAM50_Subtype,
   col = list(
    `PAM50_subtype` = subtype_colors
```

```
    ),
    annotation_name_side = "left"
)
```

```
heatmap_colors <- colorRamp2(
  c(-2,0,2),
  c("#2166AC", "white", "#B2182B")
)
```

```
gene_heatmap <- Heatmap(
  scaled_count_matrix,
  name = "Gene Expression",
  col = heatmap_colors,
  top_annotation = subtype_annotation,
  column_split = heatmap_metadata$PAM50_Subtype,
  cluster_rows = TRUE,
  cluster_columns = TRUE,
  cluster_column_slices = FALSE,
  show_column_names = FALSE,
  show_row_names = TRUE,
  row_names_side = "left",
  row_names_gp = gpar(fontsize = 8),
  row_title = "Selected Genes",
  column_title = "Samples Grouped by PAM50 Subtype",
  use_raster = FALSE,
  heatmap_legend_param = list(
    title = "Expression\nZ-score"
  )
)
```

```
draw(
  gene_heatmap,
  heatmap_legend_side = "right",
  annotation_legend_side = "right"
)
```


Samples Grouped by PAM50 Subtype
PAM50 subtype
Selected Genes
CHEK2
BRIP1
BARD1
BRCA2
RAD51D
BRCA1
RAD51C
CDK12
PALB2
ATM
Expression Z−score
2
1
0
−1
−2
PAM50 subtype
Basal
Her2−enriched
Luminal A
Luminal B
Normal